

Winterizing

Vacuum Sand Filter

Follow the suggested procedures below:

1. Lower water level in pool below stainless steel gutter.
2. Shut down auto-fill controller and domestic water supply.
3. Once the water level is lowered, close the main drain valve, perimeter overflow valve, return to pool valve to isolate the filter from the pool and open remaining valve(s) to prevent pipes breaking if freeze occurs.
 - a. In high water table areas, we recommend that some water is left in the filter to prevent lifting from hydrostatic pressure from ground water. If this is not an issue please pull water level down to the sand level, turn off pump, refer to step #3, and place a small bilge pump in the backwash trough to remove remaining water.
4. Remove drain plugs from pump box (if available) on filter tank and pump housing.
5. Drain and blow out all auxiliary lines such as chlorinators, heaters, sample lines, etc.
6. Check the filter and pipes periodically through winter months for possible excess water in these areas in case of valve leakage or rain water.
7. Cover pool if possible or remove debris ASAP to prevent surface damage or excess algae.
8. Power down Mark 5, recirculation pump and VFD (if provided). Put the VFD disconnect switch in the off position. Supplemental heat will be required if the room temperature drops below (-20° F).
9. Disconnects for recirculation pump and optional VFD should be lockout / tag out to prevent accidental powering of equipment.

Helpful Hint:

It is recommended that a manual backwash at a full flow rate for an extended time of 5 to 6 minutes is done a minimum of once a year. If your system has the optional air scour feature, it is recommended that the air scour feature is not used during this suggested preventive maintenance backwash. This suggested manual extended flow backwash extends the media life and could prevent having to replace the sand in your filter.

Vacuum Sand Filter

General Trouble Shooting

Questions to-think about when trouble shooting:

AIR BUBBLES:

1. Are unions on air scour line tight?
2. Are union nuts on ball valve tight?
3. Put second valve on air line after pump.
4. Is ball valve on air scour closed:
 - a. Is air scour valve leaking air?
5. Is MOL level set correctly?
6. Are bolts on pump suction header loose?
7. Is there a leak or restriction in suction line?
8. Is the flange connection on suction box loose?
9. Is the suction box cracked?
10. Is there a cracked flange, pipe, fitting or suction header?
11. Are air scour lines turned in correct position (up)?
12. Inspect air scour pipe and fittings for cracks and leaks.
13. Close #2 pump suction down to 18" vacuum (see if you get air – if so, possible crack in box, stainless steel suction box)
14. Check holes in screen. Add some
15. Check union under equalization screen

PUMP CAVITATES:

1. Is flow rate set? TDH set?
2. Is strainer basket sealed? (not sucking up air)
3. Is there a good seal between pump and strainer?
4. Is filter being back washed properly?
5. Is impeller worn?
6. Is the volute damaged?
7. Are valves set correctly?

Vacuum Sand Filter Trouble shooting – continued:

CLOUDY POOL WATER:

1. Has pool been backwashed properly?
2. Is the filter in filter mode (not by passing) valves 4 & 8 closed securely?
3. Is water chemistry correct? (Alkalinity – PH)?
4. How old is filter media?

REDUCED FLOW RATES:

1. Is UV strainer clean?
2. Has filter been backwashed?
3. Are all valves set correctly? (Any partially closed)?
4. Is air scour being used? (If available).
5. Are pressure gauges reading correct pressure?
6. Is the flow meter operating correctly? (Has it been cleaned)?
7. Is the pump rotation turning correct direction?
8. Are the suction or discharge line restrictions?
9. Is there Impeller blockage or obstruction?
10. Is the Impeller worn?

PUMP LOSES PRIME:

1. Is water level too low? (MOL setting)?
2. Is there an air leak on suction side of pump?
3. Is filter clean? (Has filter been backwashed?)

SAND IN POOL:

1. Was water level too high when air scour was turned on?
2. Is filter media correct size and installed with proper freeboard?
3. Does filter have broken laterals?
4. Does filter have a cracked header?

Vacuum Sand Filter Trouble shooting – continued:

HAVE TO BACKWASH TOO OFTEN:

1. Does pool have heavy bather loads?
2. Was back wash performed properly?
3. Have lotions or oils been introduced to pool?
4. Does filter media need changing?

SHUTS DOWN TO OFTEN FOR BACKWASH:

1. Is vacuum limit switch adjusted to correct setting?
2. Is vacuum limit switch bad?
3. Are valves set properly?
4. Does filter media need changing?

DOES NOT SHUTDOWN FOR BACKWASH:

1. Is vacuum limit switch adjusted to correct setting?
2. Is vacuum limit switch bad?
3. Are controls wired correctly?

PUMP MOTOR RUNNING HOT:

1. Is there low voltage coming to motor?
2. Does pump motor have enough ventilation?
3. Is strainer basket full or restricted?
4. Is TDH set correctly?
5. Is impeller binding or rubbing?

PUMP MOTOR NOISEY:

1. Does the motor have bad bearings?
2. Is the impeller rubbing?
3. Is strainer basket full or restricted?
4. Is supply or suction restricted? (Check valve settings)?

NO BUBBLES DURING AIR SCOUR:

1. Is valve open to air scour?
2. Is union connected securely?
3. Is there a blockage in the line or fittings?